

Ramya Velaga

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SUMMARY

Senior Machine Learning Engineer with 5+ years designing, implementing and scaling ranking, targeting, and optimization models for seller and buyer side ad tech at PubMatic. Improved the production LightGBM based bid ranking and throttling system that scores and paces bids under real time auction constraints and has prototyped PyTorch based embedding models to explore deep learning upgrades to that system. Has driven \$5M+ in annual revenue through end-to-end model ownership from design through production, partnering directly with customer success, engineering, and product teams to translate ML metrics into business decisions.

SKILLS

- **Programming:** Python, SQL, Scala, C++
- **Data & ML Systems:** Feature Engineering, Feature Stores, Data Pipelines, ETL, Predictive Modeling, Experimentation, Statistical Analysis
- **Data & Cloud:** Snowflake, Apache Spark, PySpark, Hadoop, AWS, DynamoDB, MongoDB
- **ML / AI:** PyTorch, Deep Learning, Embedding Models, Reinforcement Learning, LightGBM, XGBoost, Generative AI, Agentic AI
- **Ad Tech & Ranking:** Bid Ranking, Ads Targeting, Auction Optimization, Pacing, Real Time Bidding (RTB), Recommendation Systems
- **Production & MLOps:** Docker, Kubernetes, MLflow, Airflow, Git, CI/CD Pipelines, ML Observability
- **Data Processing & Analytics:** Polars, Pandas, NumPy, Data Visualization, A/B Testing, Experiment Design, Explainable AI (SHAP)

EXPERIENCE

Senior Machine Learning Engineer – PUBMATIC Inc, Redwood City, California

Sep 2025 – Present

- Improved the production bid ranking and throttling system that scores incoming bids with a LightGBM model using impression level and user level targeting features, then paces delivery against campaign QPS limits, including refactoring pipelines in Polars to cut runtime 75% and keep 1000+ demand side buyers within contractual pacing.
- Prototyped user and domain level embedding models in PyTorch to test deep learning based targeting signals as a potential upgrade path for the production ranking model.
- Recovered an estimated \$45K in daily ad spend (~\$16M annualized) by identifying and resolving a critical model inference defect through root cause analysis of production data and model behavior.
- Authored design documents and led reviews across 3+ cross functional teams, including product, engineering, and business, to align on ranking and targeting model changes and their downstream impact on buyer outcomes.
- Cut QPS debugging time by 91% and eliminated manual monitoring by building AutoTune, a three layer agentic system (deterministic health classification across 40+ metrics plus LLM based root cause analysis) running autonomously every 2 hours, unlocking \$20K/day in revenue upside.

Machine Learning Engineer – PUBMATIC Inc, Redwood City, California

June 2024 – Aug 2025

- Generated \$5K in incremental daily revenue (~\$1.8M annualized) by designing and validating the SPO Take Rate Optimization model, then rolling it out in stages with rigorous experimentation and KPI based monitoring.
- Retained an at risk demand partnership (Beeswax) by doubling fill rate and growing Share of Voice by 15K QPS, using a custom ML driven objective metric to optimize QPS allocation.
- Strengthened QPS prediction accuracy and system adaptability by fixing critical bugs, migrating configuration to a SQL database driven architecture, and refining core algorithms.

Machine Learning Engineer – PUBMATIC Inc, Pune

Jan 2022 – June 2024

- Delivered \$7.35M in incremental annual revenue by redesigning and retraining take rate optimization models to recommend optimal bid strategies under real world business constraints.
- Boosted eCPM by 10% and platform spend by \$90K per day by running offline feature analysis and A/B tests on bid throttling algorithms to validate new bidding parameters.
- Added \$1M in annual revenue by extending the take rate model to support multiple deal pricing structures, updating core model logic and the underlying data schema.
- Improved data pipeline reliability and streamlined cross team integration by redesigning PySpark calculation workflows around existing data fields.

Machine Learning Engineer – MINDBODY Inc, Pune

Aug 2020 – Jan 2022

- Built the company's first end to end feature store (Snowflake offline, DynamoDB real time) powering 10+ ML use cases across Mindbody's large scale consumer platform, cutting data processing time 50%+ and Lead Scoring delivery time in half (two quarters to one).
- Built anomaly detection models to catch outliers missed by rule-based thresholds, plus a business advisor chatbot for self service analytics.
- Grew user engagement through LightFM based course recommendations and a Similar Trainer ranking model built on user and synthetic interaction data.

PROJECTS & HACKATHONS

AI-Powered Revenue Change Analysis for Publishers, Finalist (PubMatic Hackathon)

- Built an anomaly detection framework (Prophet, ensemble Isolation Forest, Z-score analysis) to automatically flag publisher revenue changes.
- Paired it with a root cause framework (XGBoost delta analysis, SHAP feature attribution, statistical correlation) projected to address ~\$2.3M in 2024 P1 related revenue loss and save an estimated 4,320 engineering hours annually.
- Technologies: Prophet, Isolation Forest, XGBoost, SHAP, Python

Bid Density Prediction, Finalist (PubMatic Hackathon)

- Built a multi class XGBoost classifier predicting per impression bid density (0/1/2+) using targeting features (geo, platform, format, user signals) to directly inform bid ranking decisions.
- Technologies: XGBoost, LightGBM, Python

Voice AI Travel Assistant (DeepLearning.ai Voice AI Hackathon)

- Built an end-to-end voice-based travel assistant integrating flight, hotel, rental car, and payment APIs with personalized recommendations.
- Technologies: Vocal Bridge, Sabre API, PayPal Sandbox, Landing AI

AWARDS & RECOGNITION

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- Special Mention (PubMatic hackathon), AutoTune, an autonomous QPS observability & tuning agent now running in production daily.
 - Finalist (PubMatic hackathons), Bid Density Prediction (Jul 2024) and AI-Powered Revenue Change Analysis for Publishers (May 2025).
 - Received 4 recognition awards: Leaders & Innovators (Q2 2022, Q2 2023), Innovation Award (2025), Biased Toward Action (2026).
 - Ranked #1 in batch, B.Tech CS, IIT Tirupati (all 4 years), and awarded the Institute Gold Medal for top academic performance.

EDUCATION

Indian Institute of Technology, Tirupati

Bachelor of Technology in Computer Science and Engineering

Aug 2016 – May 2020

PUBLICATIONS

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- “A Platform for Large-Scale Auto-Annotation of Scanned Documents Featuring Real-Time Model Building and Model Pooling,” CVIP 2021.
 - “Offline and Online Feature Store for Faster and Consistent Machine-Learning Modelling in the Wellness Domain,” EasyChair Preprint No. 7179.